Project Title	**🍔 Online Food Delivery Analysis: Data-Driven Business Insights**
Skills take away From This Project	**Python for Data Analytics (Pandas, NumPy)** **Data Cleaning & Preprocessing** **Exploratory Data Analysis (EDA)** **Handling Noisy & Real-World Data** **Business-Oriented Data Analytics** **SQL & Relational Databases** **Power BI Dashboard Development** **KPI Design & Performance Metrics** **Data Storytelling & Visualization**
Domain	**E-Commerce** **Food & Beverage Industry** **Logistics & Supply Chain** **Customer Analytics** **Business Intelligence**

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Problem Statement:

Online food delivery platforms generate massive volumes of transactional and behavioral data related to customers, restaurants, orders, payments, and deliveries.

The goal of this project is to **analyze online food delivery data** to:

- Understand customer ordering behavior
- Identify operational inefficiencies in delivery
- Evaluate restaurant performance
- Track revenue, profit, and cancellations
- Provide **data-driven insights** for business decision-making

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The project emphasizes **real-world noisy data**, requiring thorough **EDA and preprocessing** before analytics and visualization.

Business Use Cases:

- Help food delivery companies optimize delivery time and logistics
- Identify high-performing and underperforming restaurants
- Improve customer satisfaction by reducing delays and cancellations
- Understand customer preferences by city, cuisine, and time
- Enable management to track KPIs using dashboards
- Support strategic decisions on discounts, pricing, and promotions

Approach:

🔹 Step 1: Data Collection

- Dataset consists of **100,000 online food delivery orders**

🔹 Step 2: Data Understanding

- Dataset contains **25 features** covering:
    - Customer details
    - Order information
    - Restaurant attributes
    - Delivery performance
    - Financial metrics

🔹 Step 3: Data Cleaning & Preprocessing

Key preprocessing steps:

- Handling missing values using mean, median, or mode
- Removing or capping outliers (delivery time, order value)
- Correcting invalid values (ratings > 5, negative profit margin)
- Standardizing categorical values
- Ensuring logical consistency (e.g., cancelled orders vs ratings)

🔹 Step 4: Exploratory Data Analysis (EDA)

EDA is performed to uncover hidden patterns and trends:

- Distribution of order values and delivery time
- City-wise and cuisine-wise order analysis
- Weekend vs weekday demand
- Distance vs delivery delay relationship
- Cancellation reasons analysis
- Correlation analysis among numeric features

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🔹 Step 5: Feature Engineering

Derived analytical columns:

- Order day type (Weekday / Weekend)
- Peak hour indicator
- Profit margin percentage
- Delivery performance categories
- Customer age groups

- **Step 6: Data Storage (MySQL)**
  - Cleaned data can be stored in **MySQL**
  - Table creation with proper data types
  - Data inserted using SQLAlchemy
  - Enables scalable querying and reporting

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Analyst Tasks (EDA & Analytics)

Customer & Order Analysis

1. Identify top-spending customers
2. Analyze age group vs order value
3. Weekend vs weekday order patterns

Revenue & Profit Analysis

4. Monthly revenue trends
5. Impact of discounts on profit
6. High-revenue cities and cuisines

Delivery Performance

7. Average delivery time by city
8. Distance vs delivery delay analysis

9. Delivery rating vs delivery time

Restaurant Performance

10. Top-rated restaurants
11. Cancellation rate by restaurant
12. Cuisine-wise performance

Operational Insights

13. Peak hour demand analysis
14. Payment mode preferences
15. Cancellation reason analysis

🔟 DASHBOARD - POWER-BI(OR) STREAMLIT.

- Total Orders
- Total Revenue
- Average Order Value
- Average Delivery Time
- Cancellation Rate
- Average Delivery Rating

- Profit Margin %

Results:

- A **cleaned and normalized online food delivery dataset** with handled missing values, corrected inconsistencies, and treated outliers, suitable for real-world analysis.
- A **structured analytical dataset** containing all customer, order, delivery, restaurant, and financial features along with derived metrics such as peak hours, order day type, and profit margins.
- A **set of business-focused analytical queries and EDA insights** revealing customer behavior, revenue trends, delivery performance, and cancellation patterns.
- An **interactive Power BI dashboard** showcasing key KPIs, trends, and performance metrics with filters for city, cuisine, date range, and order status.
- **Comprehensive documentation** explaining the dataset, preprocessing steps, analytical approach, insights, and business recommendations for decision-making.

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📊 Project Evaluation Metrics

- **Data Cleaning & EDA Accuracy (35%)**
  Quality and correctness of data preprocessing, handling of missing values, outlier treatment, data validation, and creation of derived analytical fields.
- **Data Analytics & Business Insights (35%)**
  Effectiveness, correctness, and depth of analytical logic used to extract customer, order, delivery, and revenue insights, including KPI formulation and interpretation.
- **Visualization & Dashboard (10%)**
  Clarity, interactivity, and analytical depth of the Power BI dashboard, including appropriate use of visuals, filters, and drill-downs.

- **Documentation & Communication (10%)**
  Clarity and completeness in explaining the dataset, EDA process, analytical findings, and business recommendations.

- **Project Organization & Reproducibility (10%)**
  Logical folder structure, naming conventions, code readability, and the ability to reproduce results using provided datasets and scripts.

Data Set:
https://drive.google.com/file/d/1ln_YbLXHZjah8c-zgrT_zacnfSwTB9ZE/view?usp=sharing

Project Deliverables:

Source Code:

- Python scripts for data extraction and cleaning.
- SQL queries for creating and populating tables in the database.

SQL Database:

- Structured and normalized database containing the cleaned data.

Power BI Reports:

- A set of interactive reports and dashboards showcasing key insights.

Documentation:

- Detailed documentation explaining the process, challenges faced, and solutions implemented.

Project Guidelines:

- Use best practices for SQL database design and normalization.
- Follow Power BI best practices for creating interactive, user-friendly dashboards.

References:

Dataset explanations	ONLINE_FOOD_DELIVERY_A…
Project Live Evaluation Metrics	Project Live Evaluation
EDA Guide	Exploratory Data Analysis (EDA) Guide
Capstone Explanation Guideline	Capstone Explanation Guideline
GitHub Reference	How to Use GitHub.pptx
HOW TO ESTABLISH SQL CONNECTION	PYTHONSQLCODE_TAMIL.ipynb
Power BI tutorial	PPB - The Ultimate Power BI
Connect MySQL to Power BI	Connect MySQL to PowerBI

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Steps to Create a Free Power BI Work Account	Steps to Create a Free Power BI Work Account
POWERBI_MYSQL_CONN_ISSUe	POWERBI_MYSQL_CONN_ISSUE
Streamlit - Tamil - Recordings	Streamlit - Tamil
Streamlit - English - Recordings	Special session for STREAMLIT
Project Orientation (Tamil)	Project Orientation Session : Onlin…
Project Orientation (English)	

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PROJECT DOUBT CLARIFICATION SESSION (PROJECT AND CLASS DOUBTS)

About Session: The Project Doubt Clarification Session is a helpful resource for resolving questions and concerns about projects and class topics. It provides support in understanding project requirements, addressing code issues, and clarifying class concepts. The session aims to enhance comprehension and provide guidance to overcome challenges effectively.

Timing: Monday-Saturday (3:00PM to 4:00PM)

Booking link : **Project Doubt Session [DS/AIML]**

LIVE EVALUATION SESSION (CAPSTONE AND FINAL PROJECT)

About Session: The Live Evaluation Session for Capstone and Final Projects allows participants to showcase their projects and receive real-time feedback for improvement. It assesses project quality and provides an opportunity for discussion and evaluation.

Note: This form will Open only on Saturday (after 2 PM) and Sunday on Every Week

Timing:
For DS and AIML
Monday-Saturday (05:00PM to 06:00PM)

- ***Booking link :*** **https://forms.gle/1m2Gsro41fLtZurRA**

Evaluation Metrics *:* **Project Live Evaluation Metrics**

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